

FIG-003. Biological breadth beyond a single use case

Stage 10.14 readable review render from the Stage 10.5 panel-evidence crosswalk

A Independent public system matrix

Purpose

Show counted source systems, biological domains, source access, and evidence roles.



Reader readout

The breadth claim rests on four counted independent public systems across at least three biological domains.

Stage 10.3 main biological breadth

B DRG calcium residence-amplitude divergence

Purpose

Use a public calcium trajectory system to show a residence-versus-amplitude comparison outside the reference use case.



Reader readout

Residence and amplitude can diverge in neuronal calcium dynamics under declared window rules.

Stage 7.3/10.3 main biological breadth

C ERK GPCR residence-amplitude divergence and sufficiency...

Purpose

Use GPCR-linked ERK dynamics to show both residence/amplitude structure and comparator-sufficient boundaries.



Reader readout

RhoDyn's breadth is strengthened by systems where residence helps and systems where simpler summaries can be sufficient.

Stage 7.3/10.4 main biological breadth

D Cell Painting and MitoTox endpoint architecture

Purpose

Bring endpoint perturbation data into the same method frame without presenting it as trajectory evidence.



Reader readout

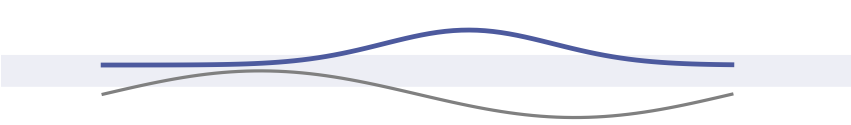
Public endpoint data can support reduced-architecture and reserve-like decisions when measurement scope is explicit.

Stage 7.4/10.3 main biological breadth

E MLCI tracking residence-amplitude stress test

Purpose

Use microbial live-cell tracking as a non-molecular trajectory stress test for schema portability and residence divergence.



Reader readout

The method can expose residence/amplitude divergence outside reporter kinase or calcium datasets, while the biological interpretation remains tracking-intensity scoped.

Stage 10.3/10.4 main biological breadth

F Source eligibility and deferred datasets

Purpose

Make source-access and license boundaries visible rather than silently adding weak demonstrations.



Reader readout

Birtwistle ERK/Akt is source-verified but deferred from counted release evidence because explicit license support is missing.

Stage 10.3 main boundary